

miRNA target-site confidence — filter-ladder plan

Goal. Turn the universal seed-scan (`utr_seed_sites.tsv.gz`, 767 arms, all seed matches with positions) into a **graded-confidence** site set by applying a **ladder of published filters**, each rung trading recall for precision. Report site/edge counts **and a validation metric at every rung** so we pick the operating point empirically rather than by fiat.

Operating principle (how we “believe it”). Each rung is a filter/score layered on the seed-scan sites. Climbing = higher precision. We validate by recomputing, at each rung, the **functional signal** on the curated $\pm$ sets — the oncomiR-TSG positive set should separate from the miRNA-poor negatives *more* as we climb (and acquired- ρ / net-repression should strengthen). If precision rises monotonically up the ladder, the filters are doing real work; the rung where it plateaus is the operating point.

All gold-standard methods are cited; thresholds follow the source papers.

Inputs (availability)

input	source	status
seed-scan sites (type + UTR position)	<code>utr_seed_sites.tsv.gz</code>	✓ built
MANE 3'UTR sequence	GENCODE v49 + GRCh38 FASTA	✓ (faidx reader)
miRNA full mature seq (for 3'-supp)	<code>mature.fa</code>	✓
conservation (PCT)	<code>Conserved_Family_Info.txt</code> (col 11 PCT, Friedman 2009)	✓ (conserved families only)
APA / UTR shortening	APAAtlas	✓
AGO-CLIP peaks / chimeric eCLIP	<code>breast_pdui_per_gene.tsv</code> POSTAR3 / ENCORI CLIP; Manakov chimeric (Zenodo 14730307)	↓ download (L5)
TarBase v9 (edge support)	DIANA Homo_sapiens_TarBase-v9. <code>tsv.gz</code>	↓ downloading (orthogonal)

The ladder (rungs, increasing precision)

L0 — Seed-match catalog (built). All canonical sites: 8mer > 7mer-m8 > 7mer-A1 > 6mer (Bartel 2009 site taxonomy). Recall ceiling; `n_6mer/n_7mer_plus/n_8mer` + positions already emitted.

L1 — Site-type stringency + weighting (Grimson 2007; Agarwal 2015). - *Filter*: drop **6mers** (~50% chance background in a 6 kb UTR; weakly functional). Keep 7mer-A1/7mer-m8/8mer. - *Weight*: per-site type coefficient = the mean context++ repression by type from Agarwal 2015 (8mer $\gg$ 7mer-m8 > 7mer-A1). Store `type_weight`.

L2 — Local context (Grimson 2007 “AU / position”; folded into Agarwal 2015 context++). - *Local AU content*: position-weighted AU fraction in the 30 nt up- and down-stream of the site (Grimson 2007 weighting $\propto 1/\text{distance}$). Higher AU $\Rightarrow$ more accessible $\Rightarrow$ functional. Store `local_AU`. - *Position-in-UTR*: (a) **exclude sites within 15 nt of the stop codon** (ribosome displacement, Grimson 2007); (b) `min_dist` = min(distance

to UTR 5' end, distance to 3' end) — sites near UTR ends score higher; deep-interior sites in long UTRs are down-weighted. *Filter*: drop the <15 nt sites; *weight*: by AU and min_dist.

L3 — 3'-supplementary pairing (Grimson 2007). - Check Watson–Crick pairing of miRNA **nt 13–16** against the UTR region 5' of the seed match at the canonical offset (seed-match 5' end – ~1..5 nt window). Flag/up-weight sites with ≥ 3 contiguous supplementary pairs (strongest for 8mer). Store `supp_pairs` (0–4) and `has_3p_supp`.

L4 — Conservation (Friedman 2009 PCT; Lewis 2005). - Join each site to `Conserved_Family_Info.txt` on (miR family, gene, UTR position) → **PCT** (probability of conserved targeting). *Filter*: $\text{PCT} \geq 0.5$ = conserved (tunable; report the PCT distribution). Defined only for conserved families (~321 arms); other arms get $\text{PCT} = \text{NA}$ → flagged **unconserved** (lower confidence, not hard-excluded, since non-conserved functional sites exist).

L5 — Experimental occupancy (gold standard). - *AGO-CLIP*: intersect `site_pos` (lifted to genomic coords) with AGO-CLIP/eCLIP **peaks** (POSTAR3 or ENCORI CLIP clusters). Site `clip_supported` if it falls under a peak. - *Chimeric eCLIP*: a site with a direct miRNA–target **chimeric duplex** read (Manakov) = `chimeric_supported` — the strongest single-site evidence ([`orphan-chimeric-eclip-validation`])). - This rung is occupancy truth, not prediction; reported as the ceiling of belief.

Definitions & clarifications (what each thing actually means)

L1 type_weight. *Not* a confidence — it's an **efficacy coefficient**: the mean repression a site of that type confers, taken from Agarwal 2015 (TargetScan7 context++ regressions): **8mer 0.31, 7mer-m8 0.161, 7mer-A1 0.099** (log-fold-change magnitude). So one 8mer $\approx$ the repression of $\sim 2 \times 7\text{mer-m8} \approx 3 \times 7\text{mer-A1}$. It lets us *weight* sites by expected potency instead of counting them equally (an 8mer is worth ~ 3 weak sites).

Why drop sites <15 nt from the stop codon (L2). The terminating ribosome covers ~the first ~15 nt of the 3'UTR; a seed match sitting in that “ribosome shadow” is **sterically occluded** — RISC can't load there while the ribosome terminates/recycles. Grimson 2007 measured it: sites within 15 nt of the stop are essentially non-functional. So it's a mechanistic accessibility filter, not statistical.

What L3 is (3'-supplementary pairing). The seed (miRNA nt 2–7/8) gives specificity; some sites get *extra* stability from the miRNA's **3' end (nt 13–16)** base-pairing to the UTR just 5' of the seed match. We revcomp nt 13–16 and look for ≥ 3 contiguous matches in a 17-nt window upstream of the seed. Sites with it are stronger (especially 8mers). It's the “does the back half of the miRNA also grab on” check.

What PCT is. Probability of Conserved Targeting (Friedman 2009; TargetScan's `Conserved_Family_Info` col). For a site, the probability that its conservation across vertebrates is due to **selection to maintain targeting** rather than background sequence conservation. $\text{PCT} \geq 0.5 \approx$ “conserved on purpose.” Defined only for the ~321 conserved miRNA families; non-conserved arms get $\text{PCT} = \text{NA}$ (flagged, not excluded — non-conserved functional sites exist).

AGO-CLIP vs chimeric eCLIP. Both crosslink Argonaute to RNA, but: - **AGO-CLIP** (POSTAR3, ENCORI CLIP) sequences the *target* fragment under AGO → tells you **where AGO sits** (an occupancy peak) but **not which miRNA** guided it; you *infer* the miRNA from the seed match under the peak. - **Chimeric eCLIP / CLASH** (Manakov) ligates the miRNA and its target into **one chimeric read** → you read **both partners directly**, so it names the exact miRNA–site duplex. Strongest single-site evidence, but sparser.

Third validation table vs L5 — different resolution. - **Third validation** (`validate_ladder_experimental.py`, edge-level): per (**miRNA, gene**) **edge**, is the *interaction* supported *anywhere*? Columns = ENCORI (AGO-CLIP interactions), TarBase v9 (validated interactions), Manakov (any chimeric duplex for the pair).

Answers “is this edge real.” - **L5** (site_genomic_l5.py, site-level): per **site position** (lifted to GRCh38), does the *predicted nucleotide location* fall under experimental binding? Columns = Manakov duplex, TarBase-CLIP coords, POSTAR3 AGO peak. Answers “is this exact site occupied.”

Why ENCORI and POSTAR-AGO are kept mutually exclusive. They are the **same assay class (AGO-CLIP)** — partly the same underlying experiments — so they are **not independent evidence**. The whole validation claim is that the three columns are *orthogonal* assays. So each table uses exactly **one** AGO-CLIP representative: ENCORI at the edge level (it ships interaction calls), POSTAR at the site level (it ships peak coordinates). Putting both in one table would double-count AGO-CLIP and fake a third independent confirmation.

Orthogonal modifiers (not rungs)

- **APA-aware UTR** (improvement #4; Mayr & Bartel 2009; APAAtlas Hong 2020). Applied **before L0** as a UTR-mask: for each gene use the APAAtlas **breast PDUI** (proximal-polyA usage). In genes with strong proximal APA (low PDUI) the distal 3'UTR is **not expressed in breast** → mask sites distal to the proximal polyA. Produces a “breast-APA-aware” site set alongside the MANE set; compare.
- **TarBase v9 edge support** (interaction-level, not site-level). Adds, per edge, whether an experiment supports it at all (and in which cell type/assay) — an orthogonal confidence axis to intersect with the site ladder.

Output

utr_site_ladder.tsv.gz — one row per **site** with: miRNA, gene, utr_pos, type, type_weight, local_AU, dist_to_stop, min_dist, position_ok, supp_pairs, PCT, conserved, clip_supported, apa_retained, and a derived **max_rung** (highest ladder rung the site survives). Plus a per-edge rollup (n_sites_Lk for k=1..5) and a per-rung summary table.

Validation plan (the belief test)

For $k = L1 \dots L5$, take the edges retaining ≥ 1 site at rung k and recompute on the curated $\pm$ sets: 1. **positive-vs-negative separation** of acquired- ρ / net-repression (Mann-Whitney; should $\uparrow$ with k); 2. **# net-repressed positives recovered** (should hold/ $\uparrow$ while negatives fall); 3. **internal consistency**: CLIP-support rate ($L5$) should rise across $L1 < L2 < L3 < L4$ (higher rungs enrich for occupied sites); 4. **APA cross-check**: APA-masked sites should be the ones with weaker breast coupling. A rung that improves (1)+(2) and is corroborated by (3) is real; the plateau = operating point.

Build order / status

1. ✓ **L1+L2+L3 built** (site_filter_ladder.py → utr_site_ladder.tsv.gz): 6,030 functional (7mer+) sites / 3,795 edges; 1,793 8mer / 2,631 7mer-m8 / 1,606 7mer-A1; 36% with 3'-supp pairing.
2. ✓ **L4 (PCT) built**: 4,100 sites carry PCT (conserved-family arms); 2,542 conserved (PCT $\geq$ 0.5), median PCT 0.68.
3. ✓ **APA modifier built**: 988 sites in breast-APA-shortened genes; 222 distal sites dropped from the parallel breast-APA-aware set.
4. ↓ **L5** — download AGO-CLIP (POSTAR3/ENCORI) + Manakov chimeric; genomic intersect. (remaining, bonus)
5. ↓ **TarBase v9** — downloading (Homo_sapiens_TarBase-v9.tsv.gz); ingest as orthogonal edge support + validation.

6. Validation harness — per-rung $\pm$ separation (pending) + TarBase enrichment.

First validation — internal coherence (no external data)

The independent rungs **agree on site strength**, which they wouldn't if they were noise:

site type	n sites	% conserved	median PCT	% 3'-supp
8mer	1,793	50%	0.75	37%
7mer-m8	2,631	40%	0.64	36%
7mer-A1	1,606	36%	0.62	34%

Rows = the **6,030 functional sites**, partitioned by seed-match type. The % **conserved column** is where the monotonic claim lives: 8mer 50% > 7mer-m8 40% > 7mer-A1 36% (L1 strength $\leftrightarrow$ L4 conservation agree). The “35% vs 26%” is the **reverse cut of the same counts — not a table row**: of the 2,542 conserved sites **35% are 8mers**, vs only **26%** of the 3,488 non-conserved (8mer share 1.35 $\times$ higher among conserved).

Definition of “conserved” (used throughout): a site is conserved iff its (miR-family, gene) carries TargetScan **PCT $\geq$ 0.5** (probability of conserved targeting, Friedman 2009). The 3,488 non-conserved are sites with PCT < 0.5 **or** — the majority — whose miRNA family isn't in TargetScan's conserved-family set at all (no PCT defined $\rightarrow$ counted as non-conserved, *not* dropped). So “conserved” is a positive-selection flag, not mere sequence identity.

Second validation — experimental AGO-CLIP (ENCORI) — PASSED, strongly

Belief test: do higher ladder rungs **predict experimental binding**? Per-edge ladder features vs ENCORI AGO-CLIP support (P(CLIP-supported | feature)):

ladder feature	n edges	% AGO-CLIP supported
weak (no 8mer / cons / 3'-supp)	888	9%
8mer (L1)	1,551	46%
3'-supp (L3)	1,748	37%
conserved PCT $\geq$ 0.5 (L4)	1,513	65%
8mer + conserved + 3'-supp (top rung)	391	74%

$\sim$ 8 $\times$ enrichment from weak $\rightarrow$ top rung. The ladder rungs predict experimental occupancy.

Reading the table: **n = edges** (unique miRNA–gene pairs; 3,795 total), **not sites** — site features are rolled up per edge (has_8mer = the edge has $\geq$ 1 8mer site, etc.). Rows **overlap** (an edge that has an 8mer *and* is conserved is counted in both) — they are feature subsets, **not a partition**; **weak** = edges with none of {8mer, conserved, 3'-supp}. The “top rung” row is the intersection (8mer $\wedge$ conserved $\wedge$ 3'-supp). *Why 3,795 and not 5,219?* the ladder only contains the **HE edges that have $\geq$ 1 functional 7mer+ site** to score; the other **1,424** HE edges have no canonical 7mer+ site at all — a **site-availability** gap, **not** a statement that they lack CLIP support. Breaking down those 1,424: only **348 (24%) even have a 6mer**; the other **1,076 have no canonical seed match at all** on the MANE 3'UTR (likely non-canonical / ORF / non-MANE-isoform sites, or indirect regulation). **Would adding 6mers help?** It would recover those 348 — but we deliberately don't: a 6mer occurs **$\sim$ 50% by chance in a multi-kb UTR** (Bartel 2009), so as a *ladder rung* it adds recall at a steep precision cost and would *flatten* the very enrichment the ladder demonstrates. Reporting the 6mer count here (as context) is harmless; using 6mers as a confidence tier is not advised.

Third validation — TRIPLE experimental cross-check (ENCORI + TarBase v9 + Manakov) — PASSED

`validate_ladder_experimental.py` → `ladder_experimental_support.tsv.gz`. Three **independent** experimental layers, all monotonic up the ladder:

ladder feature	n edges	ENCORI AGO-CLIP	TarBase v9	Manakov chimeric
weak (no 8mer/cons/3'-supp)	888	10%	49%	9%
8mer (L1)	1,551	49%	69%	22%
3'-supp (L3)	1,748	39%	67%	20%
conserved PCT $\geq$ 0.5 (L4)	1,513	69%	87%	31%
top rung (8mer+cons+3'-supp)	391	78%	88%	35%

Three orthogonal assays (AGO-CLIP occupancy, validated interactions, direct chimeric duplexes) **all** rise monotonically with the ladder. The filters are calibrated against experimental truth, not just internally coherent.

Same row convention as the second table: n = edges (miRNA–gene; the same 3,795-edge rollup, overlapping feature subsets — not a partition). **Resolution note:** this table and the second are **per-EDGE** (“is the interaction supported anywhere?”); the **first-validation** table and **L5** below are **per-SITE** (“is this exact site conserved / occupied?”). Same ladder, two resolutions.

TarBase v9 — union / intersection with the HE edge set

A **pair** = one distinct (**miRNA, gene**) **interaction** — collapsing TarBase’s many per-experiment rows down to the unique miRNA→target relationships; this is the **same unit as one of our HE edges**, so pair counts are directly comparable to edge counts.

TarBase v9: 4.72M rows → **1.31M distinct (miRNA, gene)** pairs (398k breast-tissue). Of our **5,219 HE edges**: - **HE** ∩ **TarBase** = **2,968 (57%)** experimentally validated; HE ∩ ENCORI 27%; HE ∩ Manakov 15%. - HE-only (no TarBase) = 2,251; **HE** ∪ **TarBase (within our genes)** = **171,157** — TarBase adds ~33× more candidate edges on our genes (the edge-universe expansion; ingest as its own evidence tier before using).

L5 site-level — BUILT (`site_genomic_l5.py` → `utr_site_ladder_genomic.tsv.gz`)

Lifted all 6,030 MANE-3'UTR sites to **GRCh38 genomic** (UTR exon-block walk; coordinates verified — Manakov’s PTEN 3'UTR duplexes fall inside PTEN’s GENCODE 3'UTR) and overlapped with **three independent CLIP datasets**: Manakov chimeric (miRNA-specific duplexes), TarBase-CLIP site coordinates (miRNA-specific, HITS/PAR-CLIP/CLASH), and POSTAR3 AGO-CLIP peaks (2.36M, AGO occupancy). *Does the predicted site coincide with experimental binding?* — all three climb the ladder:

rung	n	Manakov	TarBase-CLIP	POSTAR-AGO	any
7mer-A1	1,606	6%	24%	48%	55%
7mer-m8	2,631	8%	35%	50%	62%

rung	n	Manakov	TarBase-CLIP	POSTAR-AGO	any
8mer	1,793	16%	46%	55%	68%
conserved	2,542	18%	59%	63%	78%
8mer+conserved+3'-24% supp	3,602	24%	71%	70%	86%

86% of top-rung sites coincide with experimental AGO/CLIP binding — the ladder is validated at single-site resolution against three orthogonal occupancy datasets. (POSTAR3 = data/external/POSTAR/human(1).txt.gz.)

TarBase v9 at the miRTarBase HIGH-EVIDENCE level (apples-to-apples)

The 171k union was the raw (HTP-CLIP-dominated) set. Filtered to the **same evidence bar as our miRTarBase HE** (low-throughput functional: Luciferase Reporter Assay + pSILAC):

TarBase tier	rows	pairs	HE $\cap$	HE $\cup$ (our genes)
raw (all methods)	4.72M	1.31M	2,968 (57%)	171,157
non-CLIP functional	167k	117k	1,599 (31%)	21,134
LTP functional ($\approx$ miRTarBase HE)	9,819	5,256	1,135 (22%)	5,914

At matched evidence, **TarBase has 5,256 LTP-functional pairs genome-wide** (from 9,819 luciferase+pSILAC rows) — comparable in *size* to our 5,219 HE edges, but **genome-wide** (all genes). The funnel to our overlap: **5,256 genome-wide $\rightarrow$ 1,830 fall on our Hallmark genes $\rightarrow$ 1,135 match our exact arm-level edges** (the 22%). So the modest overlap is partly our Hallmark-gene scoping (most of TarBase’s 5,256 target non-Hallmark genes) and partly different literature curation. The union (within our genes) expands only to **5,914** (+~700), not 171k.

What about the other 78%? — database incompleteness, not contradiction. Of our 5,219 HE edges: **22% (1,135)** are in TarBase at matched LTP evidence; **57% (2,968)** are in TarBase at *some* evidence tier (the extra ~35% TarBase carries only via CLIP/HTP, not LTP); and **43% (2,251) are in TarBase not at all** — these come from miRTarBase’s curated LTP experiments that TarBase’s curators simply didn’t include. Both miRTarBase-HE and TarBase-LTP are **sparse samples of the same true LTP-validated interaction space** (each curates a different slice of the literature), so 22% mutual agreement is concordance between two *independent* curations — not 78% failure. The unmatched HE edges are “edges TarBase hasn’t tested”, not “edges TarBase refutes”.